What Does Repairing Choice Inconsistency Actually Buy? A Budget-Set Diagnosis

Anonymous Author(s)
Affiliation
Address
email

Abstract

Repairing an AI agent's incoherent preferences is not a new idea: inference-time layers that restore transitivity, isotonic projections onto preference-defined cones, and training-time rationality penalties have all been proposed, and several report downstream gains. What no existing result establishes is *how much* coherence is worth, because every published comparison either varies the coherence assumption by swapping in a strictly richer model class — confounding coherence with capacity — or scores the outcome with another preference judgment, or treats repair as binary. We take the economics route instead. Given an LLM agent's own choices over budget sets, we compute the minimal perturbation restoring GARP-rationalizability as a single mixed-integer program, which yields a graded, cardinal dose (an Afriat-style efficiency index) where prior work has mostly used binary cycle-counting or toggled interventions, and we apply it *post hoc* to a frozen agent, so capacity, training and policy are identical across doses by construction. Scoring the repaired sequences against a fixed, pre-registered exogenous payoff that no preference judgment enters, we trace the dose–response relationship from the raw sequence toward full rationalizability at two model scales (a 1.5B-parameter model with measured coherence headroom, and a 3B-parameter model with almost none, run as an identification control rather than a second point on one scale curve). The relationship is real, strongly monotone, and precisely estimated at both scales (Spearman $\rho = 0.756$, $p < 0.0001$, $n = 60$ at 1.5B; $\rho = 0.614$, $p = 0.0011$, $n = 25$ at 3B) and survives a targeted robustness check against the mechanical confound that larger violations simply have more room to improve. We also report two findings the design was not built to predict. First, the reciprocal-price framing manipulation that motivated the study shows no detectable CCEI shift at 1.5B in either the pilot or the corrected main measurement — two statistically indistinguishable nulls, not a sign reversal — while the output-format discard rate itself remains large enough under this manipulation to matter on its own. Second, splitting single-turn elicitation into separate sequential calls at 1.5B collapses GARP pass rate from 0.40 to 0.10 ($p = 0.0073$), with zero discard confound on either arm, moving in the *opposite* direction from an independently published single-turn-vs-multi-turn finding in the same domain and model family at a larger scale — not a replication of it. Throughout, we report Bronars power beside every efficiency index and report the discard rate as a first-class per-condition outcome rather than a silently-dropped nuisance.

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

1 Introduction

An LLM agent asked to allocate a fixed budget across two goods at varying prices, repeatedly, will frequently violate the Generalized Axiom of Revealed Preference (GARP): it prefers bundle x_1 to x_2 at one budget line and the reverse at another, in a way no monotone utility function can rationalize. A growing literature repairs exactly this kind of inconsistency in LLM systems — restoring transitivity in pairwise judgments, projecting cardinal scores onto monotone cones, penalizing incoherence during training — and several of these systems report that the repaired outputs are downstream-better than the raw ones. What none of them establishes is a *dose*: how much better, as a function of how much repair was needed, isolated from the capacity and training confounds that come from comparing a richer model class to a poorer one.

We study this question in a setting where it can be answered cleanly. An LLM agent makes a sequence of budget-constrained choices over two goods. We compute, as a single mixed-integer linear program, the minimal quantity perturbation that would make the agent's own realized choice sequence GARP-consistent — a projection applied *post hoc* to a frozen agent, so nothing about its weights, training, or decoding changes across doses. The size of that perturbation is a graded, cardinal dose. We score both the raw and the repaired sequence against a fixed, equal-weight Cobb–Douglas payoff whose optimum at each budget line is a closed-form function of prices and income alone — never of the agent's own choices — so the outcome measure is exogenous to the preference data by construction, not a restatement of coherence. Tracing dose against Δpayoff gives the relationship this paper reports.

We do not claim any of the individual pieces are new; §2 concedes exactly what is not. What has not been done is the conjunction: an agent's own realized choices, a graded coherence-indexed dose, and an outcome that is not itself a preference judgment, run together so that the identification is clean. We run this at a model scale where a pilot study found measurable coherence headroom (1.5B parameters), and repeat the identical pipeline at a scale with almost none (3B parameters) as an identification control on the method itself, not as a second point on a shared dose curve across scale.

Two results were not predicted by the design. The reciprocal-price framing manipulation intended to induce coherence variation at 1.5B produced, in a pilot run, a large apparent effect that disappears once discard-selection is corrected with a capped retry protocol: the pilot's naive estimate and the corrected main-experiment estimate are both statistically indistinguishable from zero, and the two should not be read as a sign reversal of a real effect. We report the discard rate itself, not the CCEI point estimate, as the more reliable signal that the manipulation disrupted the agent (our claim C3). Wang et al. [2025a] already recommend reporting invalid-response proportions as a checklist item; our contribution is a quantitative demonstration of how much a naive discard rule can distort a measured effect at this specific scale, broken down by attempt outcome in §5.2, not the recommendation itself. Separately, splitting single-turn elicitation into separate sequential calls at 1.5B produces a large GARP-pass-rate collapse, moving in the *opposite* direction from an independently published single-turn-vs-multi-turn finding in the same domain and model family at a larger scale, with no discard confound on either arm.

Contributions. (1) An application — not a novel method — of Demuynck and Rehbeck [2023]'s minimal-quantity-error MILP projection to an agent's own realized choices, independently verified on every output rather than trusted from solver status alone. (2) A pre-registered, closed-form exogenous payoff audited adversarially for leakage and for the mechanical confound that larger violations simply have more room to improve. (3) A measured dose–response relationship at two model scales, run as headroom model and null-effect identification control rather than two points on one curve. (4) An instrument-validity finding (discard-selection bias under a disruptive manipulation), corrected for by a stated retry protocol and reported as a first-class outcome. (5) A single-turn-vs-multi-turn elicitation-format manipulation that produces a large GARP-pass-rate collapse at 1.5B, in the *opposite* direction from an independently published single-turn-vs-multi-turn finding in the same budget-allocation domain and the same model family (Qwen2.5) at a larger scale.

2 Related Work

2.1 Repairing LLM preference and judgment consistency

Repairing an AI system's incoherent preferences is not new; we concede what is occupied before claiming what is not. At least six published systems restore some consistency property of an LLM's choices or judgments, three at inference time. None applies such an operator to an agent's *own* choice sequence over budget sets, indexes it by a graded coherence measure, and scores the result against a payoff into which no preference judgment enters — that conjunction is our claim (Table 2, Appendix A).

The sharpest vocabulary collision is POISE [Wang et al., 2026]: it reads pairwise labels as a fixed partial order and returns the pool-adjacent-violators projection onto the closed convex chain-monotone cone $\{s' : s'_1 \leq \cdots \leq s'_m\}$, with a proved Pythagorean guarantee that the edit lands weakly closer to a posited ground truth. We concede the minimum-distance priority unqualified. **The difference is in what can be guaranteed, not in what is computed.** Projection onto a closed convex set is non-expansive in L_2, which is what licenses that guarantee; the GARP-consistent set is a union of polyhedra, one per admissible ordering, hence not convex, so no analogue transfers. POISE also projects a *teacher's* offline training labels rather than an agent's own choices, excludes cycles by precondition, and validates on a 68-comparison human preference vote — and its own ablation shows the projection alone *lowering* overall quality by 0.50 while raising consistency by 2.27.

Three further inference-time repairs, each on a different object. Chadwick et al. [2025] project incoherent probability judgments onto the nearest coherent point by quadratic program and break intransitive preferences via a polynomial-time Kemeny approximation, nearest to ours in spirit but tested only on synthetic election data with no downstream task-quality comparison. TrustJudge [Wang et al., 2025b] cuts a judge's pairwise transitivity violations from 15.22% to 4.40%, but repairs judgments of third-party responses with no degree parameter. CONSISTRE [Sun, 2026] enforces transitivity/symmetry/functional-uniqueness over relation-extraction triples, with no budget sets and no Afriat machinery. One training-time system alters the agent itself, so it does not hold capacity fixed across doses: Buchanan and Foster [2026] fine-tune with IIA-invariance losses (compliance $0.920 \to 0.948$) with no downstream evaluation. A further training-time system is not an LLM agent at all: Aguiar and Kashaev [2026] fine-tune Chronos-2, a time-series forecasting model, on synthetic utility-maximizing-agent data to predict *human consumers'* own choices — not a coherence-repair operator on an agent's own behaviour, but a forecasting model partly trained on GARP-consistent synthetic data, with a downstream evaluation (17–31% bundle-prediction error reduction versus zero-shot) that neither of the other two training-time systems above reports.

2.2 Coherence versus downstream competence: the open sign

A parallel line varies the transitivity of the *preference model* rather than an agent's choices. GPM [Zhang et al., 2025] replaces the scalar Bradley–Terry reward head with a skew-symmetric embedding able to represent cycles; HRC/DSPPO [Huang et al., 2026] decomposes the preference function into transitive and cyclic components and schedules their weight through self-play. We cite both as friendly precedent, not rivals: in both, the cycle-tolerant arm is a strict superset model class (each states Bradley–Terry is its dimension-one special case), so coherence is confounded with capacity by construction, and on GPM's own length-controlled metric that arm *loses* in 18 of 24 head-to-head cells.

HRC/DSPPO already publishes a dose–response curve, and we say so before claiming anything about grading. Its Appendix C.4 traces nine settings of a schedule weight $\lambda \in \{-2, \ldots, 2\}$ against win rate: an inverted U, interior optimum at $\lambda = +1.0$, span 4.63 points. Four differences separate it from ours. *Object*: λ weights a component of a learned third-party preference proxy during training; ours acts post hoc on the agent's own realized choices, so capacity and training are identical across doses by construction. *Units*: λ has no reading as incoherence removed, where our projection-distance dose does. *Outcome*: every downstream number in both ICML papers is an LLM-judge score; ours is exogenous. *Endpoints*: their schedules never reach either extreme; ours runs from the raw sequence to full rationalizability.

Our closest theoretical neighbour is Andrews [2026], who argues representation theorems furnish label-free evaluation/regularization signals and proposes $1 - \mathrm{CCEI}$ as a penalty, running no experiments and proposing no inference-time operator. He states plainly — in the abstract, twice in §1, and as his first limitation — that coherence is *not* sufficient for good behaviour; that position is argued a priori and never measured, and he never asks whether *imposing* coherence helps or hurts. Ours is the empirical counterpart to a theoretical proposal that has circulated unrun. One study already reports the qualitative shape we look for: Ouyang et al. [2025] find alignment fine-tuning's relationship to realized capital-expenditure prediction is non-monotonic — moderate single-dimension alignment *raises* the predictive coefficient while full composite alignment degrades it — for a different enforced property and no minimum-distance objective, but the shape our design is built to detect.

2.3 Reliability of the instrument, and a record of repair that failed

The sharpest objection is psychometric, and it is in *PNAS*. Across eight datasets and $> 1{,}600$ participants, Nitsch et al. [2022] find not one of $\sim$40 ICC estimates for CCEI or Houtman–Maks reaches 0.75, and that 97 participants given the chance to revise their own inconsistent budget-set choices saw mean CCEI *fall*, with test–retest reliability dropping from 0.522 to 0.443. Our answer has three parts, the third a concession. First, *that revision arm is not a repair operator*: participants saw a random subset of choices, not the violating ones, with no consistency objective and no guarantee the revised set had fewer violations — our operator attains rationalizability by construction and verifies it independently. The same distinction disposes of Yamin et al. [2026]'s isotonic-calibration repair, worse in 14 of 16 cells: it projects onto the monotone *calibration* cone while the metric it is graded on scores conditional independence, a different set than the one repaired. Second, the reliability finding is diagnosed by its own authors as a between-subject-variance problem, not measurement error (within-subject CV $\approx 15\%$), with their own prescribed remedy — "a manipulation (i.e., a between-groups design)" — being this paper's design. Third, the concession, stated plainly: *there is no answer to the third-negative-in-a-row problem.* Adding Zhu et al. [2025]'s frozen-embedding probability-axiom enforcement (coherence improved, held-out MSE slightly worse), three independent results across three axiom systems point one way, and a null result here would be a fourth confirmation rather than a discovery. We therefore carry a distance-matched null-operator control — identical displacement, no consistency gain — which neither published negative had.

2.4 Economics-side interventions on LLM choice behaviour

The closest neighbour on the economics side is invisible to any arXiv sweep. Cook et al. [2026] elicit economic choices from ten open-weight models and steer them toward payoff-maximising behaviour via personas, prompt masking, and learned control vectors — occupying two of our three legs (intervention on an agent's own economic choices, continuously graded strength) without Afriat machinery, a minimum-perturbation objective, or a payoff-traced frontier; their finding that reframing and control vectors move models while persona prompting does not corroborates our manipulation choice. That choice is anchored in Wang et al. [2025a], who find *format*, not persona or temperature, moves the Afriat index in the same budget-allocation domain we study, on Qwen2.5-7B-Instruct among other models — the same model family as our 1.5B, at a larger scale: collapsing their *multi-turn baseline* to single-turn drops Qwen2.5-7B's CCEI by up to 0.241. We adopted format as an experimental arm on that evidence. Our arm runs the opposite manipulation — our own baseline is single-turn, and we split it into separate sequential calls — and finds the opposite direction: GARP pass rate collapses from 0.40 to 0.10 ($p = 0.0073$; survives Benjamini–Hochberg correction across our full test family but not the more conservative Holm correction) with zero discards. This is not a replication of Wang et al. [2025a]'s finding: in the same domain and the same model family, which elicitation format is more coherence-degrading reverses between their 7B model and our 1.5B model. We report the disagreement rather than the confirmation we expected.

3 Method

3.1 The projection: Demuynck & Rehbeck's (2023) minimal-quantity-error MILP, applied and independently verified

This section applies, rather than proposes, a projection method. The MILP formulation below is Demuynck and Rehbeck [2023]'s multiplier-free ordinal characterization of minimal-quantity-error GARP repair, adopted here on an LLM agent's own choice sequence and verified independently on every output; nothing about the formulation itself is new to this paper.

For a sequence of T observations $(p_t, x_t)_{t=1}^T$, K goods, GARP holds if there is no sequence of choices such that x_{t_1} is revealed preferred to x_{t_2}, ..., x_{t_n} is revealed preferred to x_{t_1}, with at least one strict revealed preference in the cycle. Given a GARP-violating sequence, we seek the minimal perturbation $\tilde{x}_t$ that restores GARP-consistency. The naive formulation parameterizes this with Afriat multipliers, which makes the resulting constraints bilinear in the unknowns once $\tilde{x}$ is itself a decision variable — fixing a candidate preference ordering does not remove the bilinearity, since a price-weighted multiplier term remains a product of two unknowns regardless of ordering. We instead use the multiplier-free ordinal characterization of Demuynck and Rehbeck [2023], under which every constraint is linear jointly in the perturbed bundles, a set of ordinal utility levels $u_t \in [0, 1]$ that carry no multipliers, and binary comparison indicators $U_{t,v} \in \{0, 1\}$ for $t \neq v$. At $T = 25$ this is 600 binaries and 125 continuous variables per trace, solved as a single MILP on the HiGHS backend, with no outer search over preference orderings and no alternating scheme.

Three method commitments, each carrying an explicit risk. **Budget exhaustion is imposed as an equality**, $p_t \cdot \tilde{x}_t = I_t$ for every observation (income fixed at 100 throughout this project); this is standard for the design, and it makes the big-M constant in the ordering constraints computable a priori as $\alpha > \max_t I_t$. **A fixed strict-preference margin** $\gamma > 0$, set to $10^{-4} \cdot \min_t I_t$, converts an unattained infimum into an attained minimum: the GARP-consistent set is not closed, so an unregularized projection can read a distance of exactly zero on genuinely violating data. We verified the reported distance is stable across $\gamma \in \{10^{-2}, \ldots, 10^{-6}\}$ before treating it as reportable. **Every returned $\tilde{x}$ is verified independently** via the same combinatorial Warshall-closure GARP check used everywhere else in this project, never trusted from solver optimality status alone — all 85 GARP-violating traces' projections in the main experiment were independently re-verified GARP-consistent, with MIP gap $\leq 8.1 \times 10^{-5}$ on every solve.

The objective is L_1: $\min \sum_t \sum_k w_{t,k} d_{t,k}$ subject to $\tilde{x}_{t,k} - x_{t,k} \leq d_{t,k}$ and $x_{t,k} - \tilde{x}_{t,k} \leq d_{t,k}$, with uniform weights $w = 1$. This keeps the program a pure MILP; L_∞ is reported as a free-to-add robustness check. L_2, the literature's own least-squares index, would require an MIQP solver we do not have configured and is not computed. The *dose* for a trace is its L_1 projection distance $\sum_t \sum_k |x_{t,k} - \tilde{x}_{t,k}|$, comparable across sessions without further normalization because income is fixed throughout. A Cobb–Douglas demand share-fitted to each trace's own observed data is computed as a feasibility incumbent and sanity ceiling on the reported distance — but this share-fitted quantity is never actually passed to the solver as a warm start (the underlying solver has no incumbent-injection interface), so it never influences the direction or magnitude of any reported projection; it is used only as a logged upper-bound diagnostic.

3.2 What can be guaranteed, and what cannot: positioning against convex projection

§2 positions this projection against the closest published structural analogue, a proved-optimal projection onto a convex monotone cone [Wang et al., 2026]: because the GARP-consistent set is a union of polyhedra rather than a single convex set, the non-expansiveness that licenses a weakly-closer-to-ground-truth guarantee there has no analogue here. Concretely for this formulation, the prior operator's cone is fixed by an ordering supplied as input, whereas a GARP repair must decide which revealed-preference comparisons to give up — the ordering search is absorbed into the binary comparison indicators of §3.1 rather than eliminated, which is why no distance-minimization guarantee analogous to the convex case is claimed.

3.3 The exogenous payoff

The payoff must not be derived from the agent's own revealed choices — a utility function fit to the agent's choices (such as the projection's own feasibility incumbent above) would be circular if used as the outcome measure, rewarding the agent for resembling its own revealed preferences rather than measuring anything external to them. We instead fix, before any model was queried and never re-estimated from any agent's choices, an equal-weight Cobb–Douglas valuation $U_{\text{exo}}(x) = x_A^{0.5} x_B^{0.5}$. At budget line (p_t, I_t) the optimal bundle x_t^* is the closed-form Cobb–Douglas demand $x_{t,A}^* = 0.5 I_t / p_{t,A}$, $x_{t,B}^* = 0.5 I_t / p_{t,B}$ — a function of prices and income alone, never of the agent's observed or projected bundle. The payoff score is the efficiency ratio $\text{payoff}(x_t) = U_{\text{exo}}(x_t) / U_{\text{exo}}(x_t^*) \in (0, 1]$, computed identically for the raw and projected sequence at the same budget lines, so that $\Delta\text{payoff} = \text{payoff}(\tilde{x}) - \text{payoff}(x)$ is comparable across conditions, models, and repairs. This mirrors — not the specifics, but the structure of — a money-metric utility index, the standard device for welfare comparison over the same GARP/Afriat objects this paper's method already relies on.

We audited this design adversarially, before treating any dose–response result as reportable, for four failure modes: a shared-data leak between payoff and projection, a mechanical link between GARP-consistency and payoff, a projection direction secretly aimed at the payoff optimum, and the confound that larger-dose traces simply start worse and have more room to improve. None survived the audit (Appendix B gives every check and number); the result we report in §5 is not an artifact of any of the four.

4 Experimental design

4.1 Models and conditions

Two locally-hosted open-weight models, run entirely on local compute at zero API cost: `qwen2.5:1.5b-instruct` (the headroom model, per a pilot study that found measurable within-condition coherence variation at this scale) and `llama3.2:3b-instruct` (the null-effect control, per the same pilot finding almost no coherence headroom at this scale — mean baseline CCEI 0.99). A third model family was tested for feasibility but excluded after a reproducible multi-model-residency crash under sustained rotation on the local machine; the two-model, model-major design (all sessions for one model complete before the next model loads) is both a wall-clock optimization and, after that observed failure, a correctness requirement. **Compute.** All model inference ran locally via 4-bit-quantized weights on a single consumer CPU-class machine, no GPU and no commercial API calls; the full $N = 30$ design across both models and every condition completed in approximately 2.1 hours wall-clock. Every MILP projection (§3.1) solved in under 5 seconds on the same machine, CPU-only, with MIP gap $\leq 8.1 \times 10^{-5}$ on every solve.

Three conditions at 1.5B, two at 3B. **Baseline**: direct per-token-return framing, single turn, all 25 rounds in one prompt and one response. **Reciprocal**: identical single-turn format with an inverted-price framing manipulation, the strongest manipulation identified in the pilot. **Multiturn** (1.5B only): baseline framing, but each of the 25 rounds is a separate sequential API call carrying the running conversation history, isolating the elicitation-format mechanism from the price-framing mechanism; the arm is motivated by an independently published finding, in the same budget-allocation domain we study, that collapsing multi-turn elicitation to single-turn moves CCEI by up to -0.241 for a same-family model at a larger scale [Wang et al., 2025a], an order of magnitude larger than this project's own pilot framing effect. Our arm tests the manipulation in the opposite direction (single-turn baseline split into multi-turn), since that is the direction our own baseline format already uses. It is not added at 3B, whose role is a null-effect control on the identification strategy rather than a second scale point requiring the full condition set.

4.2 Budget sets and power

Each of $N = 30$ independent replicates per (model, condition) cell draws its own fresh $T = 25$, $K = 2$ budget set (continuous-density prices, $M, N \sim U[0.1, 1.0]$ i.i.d. rejected unless $\max(M, N) \geq 0.5$, budget exhaustion enforced), seeded independently from every other replicate — unlike a pilot study's single shared set, needed because the main experiment targets independent between-condition replication, not test–retest reliability. Continuous, independently-drawn prices with enforced budget exhaustion also avoid a known pathology: CCEI can read exactly 1.0 on data that violates GARP whenever a comparison lands on exact budget-line equality, an event common under grid-sampled or rounded prices [Andrews, 2026]. $N = 30$ gives power ≥ 0.999 against the pilot's full framing-effect magnitude on the pass-rate metric and ≥ 0.80 down to $\sim$60% attenuation, but is explicitly underpowered for a CCEI-scale effect of the pilot's magnitude at 1.5B (minimum detectable ≈ 0.11), which we report rather than paper over. Every replicate's Bronars power is logged; all 150 draws held power ≥ 0.998.

4.3 The discard-selection problem, and its correction as a stated contribution

A pilot run found that under reciprocal framing at 1.5B, 52% of sessions (13 of 25) failed the required output-format contract and were silently discarded under the pilot's own naive handling — standard practice, and precisely the practice that biases a measured coherence effect toward zero, since the

Table 1: Headline results per (model, condition) cell. GARP pass and CCEI are computed on kept traces only; dose and Δpayoff are computed on the subset of kept traces that violate GARP.

Model	Condition	n kept	mean CCEI	GARP pass	mean dose (L_1)	mean Δpayoff
llama3.2:3b	baseline	30	0.9900	0.73	5.01	+0.0018
llama3.2:3b	reciprocal	28	0.9713	0.39	6.27	+0.0016
qwen2.5:1.5b	baseline	30	0.9522	0.40	16.68	+0.0090
qwen2.5:1.5b	multiturn	30	0.9540	0.10	14.54	+0.0083
qwen2.5:1.5b	reciprocal	24	0.9413	0.38	12.02	+0.0065

discarded sessions are exactly those where the manipulation disrupted the agent most. We treat this as a claim about instrument validity in its own right, not a footnote to the dose–response result, true regardless of whether projection helps, hurts, or does nothing downstream. The main experiment implements a capped retry protocol: a session failing the $\geq 20/25$-valid-rounds threshold is retried up to three total attempts with a fresh seed offset, every attempt logged; a slot still failing is a residual discard, excluded from CCEI/GARP but retained for audit. We report first-attempt and residual post-retry discard rates as a first-class per-condition outcome for every cell.

5 Results

187 attempt-records were logged across 150 replicate slots (2 models $\times$ up to 3 conditions $\times$ 30 replicates); 142 slots kept a usable trace, 8 were residual discards after three attempts. Every cell reached its full 30 replicates.

5.1 The dose–response relationship (C1)

Across all 85 GARP-violating traces with a computed, independently-verified projection (60 at 1.5B, 25 at 3B): Spearman $\rho = 0.729$ ($p < 0.0001$), Pearson $r = 0.821$ ($p < 0.0001$), mean Δpayoff $+0.0091$ (one-sample $t = 5.41$, $p < 0.00001$), with 70 of 85 traces (82%) improving after repair and 15 (18%) worsening. Figure 1 shows the relationship split by model on shared axes. **Both scales show a significant, positive relationship, and it is stronger at the headroom model**: $\rho = 0.756$ ($p < 0.0001$, $n = 60$) at 1.5B versus $\rho = 0.614$ ($p = 0.0011$, $n = 25$) at 3B. This is the one place our data contradicts our own pre-registered expectation for the 3B control, and we report the contradiction rather than silently absorb it: we expected the 3B arm to fail to reject the null, on the reasoning that a model with almost no baseline incoherence has little for the identification strategy to act on. Instead, on the smaller set of violations that 3B does produce (mostly induced by reciprocal framing, §5.2), projection toward GARP-consistency still measurably helps on the exogenous payoff, at roughly a third the strength of the 1.5B relationship. The correct reading is not that the control found nothing, but that it found a real effect of about a third the magnitude — evidence that the identification strategy is not an artifact specific to the 1.5B model, though not the null result our design anticipated.

We tested the mechanical confound that larger-dose traces might simply start from worse raw payoff and therefore have more room to improve. The confound is real (dose vs. raw payoff, Pearson $r = -0.41$, $p < 0.0001$) but does not explain the relationship away: the partial correlation of dose against Δpayoff controlling for raw payoff is $r = 0.784$ ($p = 7 \times 10^{-19}$), and an OLS regression of Δpayoff on dose and raw payoff jointly finds dose highly significant ($t = 11.4$, $p < 10^{-16}$) with raw payoff's own marginal contribution not distinguishable from zero ($p = 0.18$) once dose is included. Full detail in Appendix B.

5.2 The reciprocal-framing manipulation: a survivorship artifact, corrected

At 3B, reciprocal framing replicates the pilot's finding on an independently-drawn budget-set design: GARP pass rate collapses from 0.73 (22/30) to 0.39 (11/28), $p = 0.0089$ (survives Benjamini–Hochberg correction across our full test family, $p_{\mathrm{BH}} = 0.011$, but not the more conservative Holm correction, $p_{\mathrm{Holm}} = 0.062$); CCEI moves much less and reaches only borderline significance (0.9900 vs. 0.9713, t-test $p = 0.0508$) — the same near-1 compression the pilot flagged, and a further case

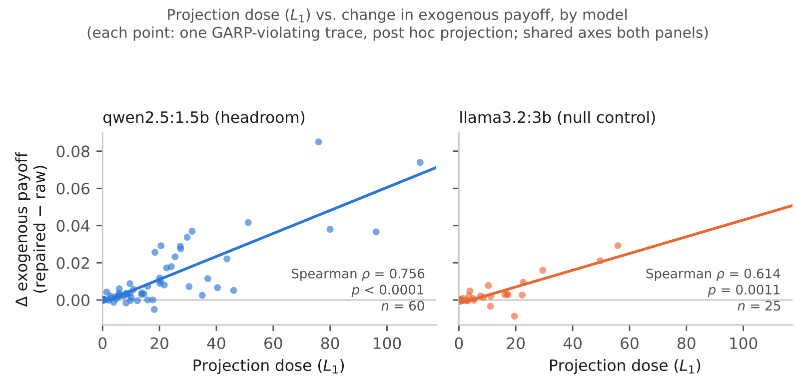

Figure 1: Projection dose (L_1) against the change in exogenous payoff after repair, for every GARP-violating trace, split by model on shared axes so the two panels' slopes are directly comparable. Both relationships are positive and significant; the headroom model's (left) is steeper and tighter than the null-control's (right), though the control's relationship is itself significant rather than null.

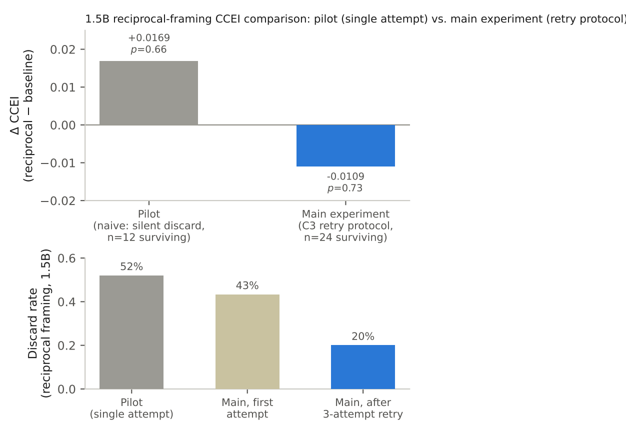

Figure 2: Top: the 1.5B reciprocal-framing CCEI comparison, pilot (naive single-attempt discard handling, grey) vs. main experiment (capped-retry-corrected, blue). Bottom: the corresponding discard rate at each stage. As it falls, the apparent CCEI "effect" collapses toward zero and reverses sign — the pilot's finding was driven by which sessions survived, not by choices changing.

in this project of the pattern that GARP pass rate is the sensitive instrument for framing/format disruption while CCEI understates it.

At 1.5B, the manipulation this study was designed around does *not* survive proper correction for discard-selection. First-attempt discard under reciprocal framing at 1.5B was 43.3% (13/30), falling to a residual 20.0% (6/30) after the retry protocol — smaller than the pilot's unretried 52%, plausibly because independent per-replicate budget sets less often reproduce one unusually hard draw, but the pattern is unchanged: reciprocal framing produces far more discards than any other condition, including the *other* 1.5B manipulation (multiturn, 0%). Once the surviving 24 sessions are compared, GARP pass rate (0.40 vs. 0.38, $p = 0.85$) and CCEI (0.9522 vs. 0.9413, $p = 0.73$) are indistinguishable between baseline and reciprocal framing; the CCEI point estimate moves in the pilot's predicted direction but is small relative to 1.5B's own noise, with confidence intervals overlapping almost entirely. The pilot's confounded $+0.0169$ finding is superseded by this properly-powered, still-noisy null, not confirmed or refuted. Figure 2 shows why: as the residual discard rate falls from 52% toward 20%, the naive framing "effect" collapses toward zero and changes sign.

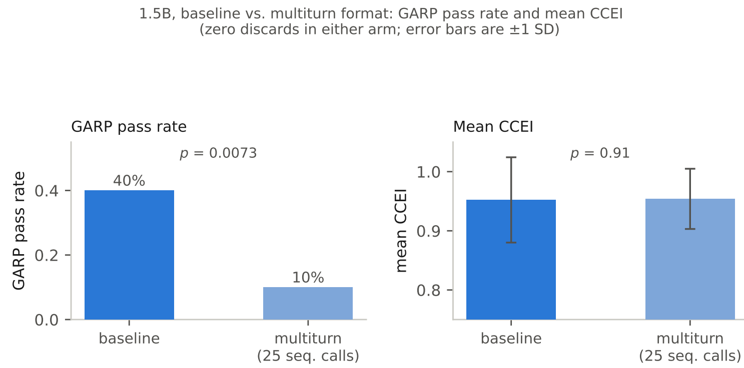

Figure 3: 1.5B, baseline vs. multiturn elicitation format. GARP pass rate (left) collapses; mean CCEI (right, ± 1 SD) is visually flat with heavily overlapping dispersion. Zero discards occurred in either arm, unlike Figure 2.

5.3 The multiturn/format effect: a large effect in the opposite direction from the literature

Splitting the 25 rounds into separate sequential calls collapses the GARP pass rate from 40% to 10% ($p = 0.0073$) while CCEI barely moves (0.9522 vs. 0.9540, $p = 0.91$; Figure 3) — a larger drop in percentage points than reciprocal framing produced at 1.5B, with *zero discards on either arm* so no selection-bias caveat applies. Corrected for multiple comparisons across our full test family, this result survives Benjamini–Hochberg correction ($p_{\mathrm{BH}} = 0.0094$) but not the more conservative Holm correction ($p_{\mathrm{Holm}} = 0.058$); we report both rather than the raw p-value alone. Together with the 3B reciprocal-framing result (§5.2), this is a further case of the pattern already established in this project: GARP pass rate is the sensitive instrument for framing/format disruption; CCEI is not. Against the literature this arm was designed against [Wang et al., 2025a]: in the same budget-allocation domain, on the same model family at a larger scale (Qwen2.5-7B), single-turn elicitation was found to be *more* coherence-degrading than multi-turn; here, at 1.5B, the reverse holds — multi-turn elicitation collapses GARP pass rate relative to our single-turn baseline. We do not have an explanation for the reversal and do not offer one; we report the disagreement as a finding in its own right, not a confirmation of the literature's own strongest lever.

6 Limitations

CCEI is noisy at 1.5B, and the paper leads with GARP pass rate because of it, not by preference. The 1.5B model's within-condition CCEI noise (baseline within-subject coefficient of variation 14.8%) is comparable to published human test–retest noise on the same index [Nitsch et al., 2022], and $N = 30$ is explicitly underpowered for a CCEI-scale effect of the pilot's own magnitude at this scale (the minimum reliably detectable effect at this N is roughly $2.5\times$ the pilot's magnitude). A CCEI-power-matched design would need $N \approx 111$ to 161 depending on the assumed baseline variance; that scale-up was deferred by the operator pending this run's results and was not executed. We treat this as an open design tradeoff, not a result masked by a metric choice: the pass-rate metric is adequately powered throughout and every CCEI result we report is accompanied by its own confidence interval rather than presented as equally reliable.

Single run per condition per model at $N = 30$. Every (model, condition) cell in this paper is one run of the stated protocol; we do not have repeated full runs of the entire pipeline to report a between-run variance on the headline dose–response statistic itself, only the within-run replicate variance captured by the reported confidence intervals and p-values.

Two model families, one scale point each. The headroom/null-control design isolates the identification strategy from a scale confound but does not trace a dose–response relationship *across* model scale: a pilot found headroom and measurement reliability do not coexist anywhere in the range piloted, so the design targets the scale where headroom exists rather than blending a noisy-but-real candidate effect with a clean-but-absent one across a wider range. A third model family was excluded after an observed multi-model-residency failure on the local machine; extending the roster is a documented follow-up, not a silent omission.

One fixed exogenous payoff. U_{exo} uses equal Cobb–Douglas weights, chosen for being a zero-degrees-of-freedom, closed-form, no-solver-dependency reference rather than swept over alternative fixed weightings; §3.3's audit addresses whether this specific choice leaks information from the agent's revealed preferences, not whether a different fixed weighting would trace the same curve.

An adverse prior the paper does not shed. Three independently published repair attempts moved the wrong way (§2). Our own reciprocal-framing manipulation at 1.5B (§5.2) is a fourth negative in the narrower sense that the manipulation intended to induce measurable coherence variation did not do so once properly measured. The dose–response relationship we do report (§5) argues against reading this prior as universal, but it is a relationship over naturally- and framing-induced violations pooled together, not a demonstration that any one manipulation reliably induces the violations repaired.

Broader impacts. A positive dose–response relationship, read carelessly, could be taken as license to impose coherence-repair on deployed AI economic agents by default. Our own result argues against that reading: 18% of repairs in this study *worsened* the exogenous payoff (§5), and the reciprocal-framing manipulation this study was designed around turned out, once corrected, to have no measurable effect on the coherence it was thought to disrupt (§5.2). The paper's contribution is a measurement design for asking whether repair helps in a given setting, not a recommendation to deploy it universally. Separately, the discard-selection finding (§4.3) generalizes beyond this project: any revealed-preference or consistency instrument that silently drops unparseable or disruption-caused failures risks systematically underestimating exactly the manipulations that matter most, which bears on how AI economic-behavior audits are designed and reported more broadly. We see no elevated misuse risk specific to this work: it uses only synthetic budget-allocation tasks on publicly available open-weight models, releases no new model or dataset, and does not involve human subjects.

7 Conclusion

Applying Demuynck and Rehbeck [2023]'s minimal-quantity-error GARP repair to an LLM agent's own realized choices, post hoc and scored against a fixed exogenous payoff, we find a real, precisely estimated, monotone dose–response relationship between repair size and payoff gain, at a model scale with measured coherence headroom and one without. The relationship survives an adversarial audit for leakage and for the confound that larger repairs simply start from more room to improve. Two unpredicted findings — a purpose-built framing manipulation's apparent 1.5B effect was a discard-selection artifact rather than a real coherence shift, and an elicitation-format manipulation produces a large GARP-pass-rate collapse in the opposite direction from an independently published finding in the same domain and model family at a larger scale — are reported in their own right: an instrument that silently discards its most-disrupted observations, and a disagreement with the closest published comparison, are exactly the kind of failure and complication this literature should surface by default rather than paper over.

References

Victor H. Aguiar and Nail Kashaev. GARP-EFM: Improving foundation models with revealed preference structure. *arXiv preprint arXiv:2603.23993*, 2026.

Andrews. Revealed rationality: Label-free evaluation and regularization from representation theorems. *arXiv preprint arXiv:2608.05015*, 2026.

Joy Buchanan and Joshua Foster. The innate economic preferences of language models. *arXiv preprint arXiv:2607.26288*, 2026.

Alina Chadwick, Anson Kahng, and Jens Kipper. Dutch books and money pumps: rectifying vulnerabilities in LLMs through rationality. In *Proceedings of the 5th International Conference on Human and Artificial Rationality (HAR)*, 2025.

Trevor Cook, Sophia Kazinnik, Alexander Modig, and Sam Palmer. What do LLMs want? Technical Report FEDS 2026-006, Federal Reserve Board, 2026.

Thomas Demuynck and John Rehbeck. Computing revealed preference goodness-of-fit measures with integer programming. *Economic Theory*, 2023.

Huang, Li, Zhao, and Li. Transitivity meets cyclicity: Explicit preference decomposition for dynamic llm alignment. In *International Conference on Machine Learning (ICML)*, 2026. arXiv:2605.17342.

Felix J. Nitsch, Lisa M. Lüpken, Nils Lüschow, and Tobias Kalenscher. On the reliability of individual economic rationality measurements. *Proceedings of the National Academy of Sciences*, 119(31): e2202070119, 2022.

Shumiao Ouyang, Hayong Yun, and Xingjian Zheng. AI as decision-maker: Ethics and risk preferences of LLMs. *arXiv preprint arXiv:2406.01168*, 2025.

Mingxuan Sun. CONSISTRE: A unified consistency-aware framework for document-level relation extraction with large language models. *arXiv preprint arXiv:2607.24312*, 2026.

Wang, Yao, Zhang, Gai, Liu, and Zhong. When experimental economics meets large language models: Tactics with evidence. *arXiv preprint arXiv:2505.21371*, 2025a.

Wang, Zhan, et al. Trustroboreward: Preference-ordered isotonic score editing for multi-paradigm robot reward modeling. *arXiv preprint arXiv:2608.08491*, 2026.

Yidong Wang et al. Trustjudge: Inconsistencies of LLM-as-a-judge and how to alleviate them. *arXiv preprint arXiv:2509.21117*, 2025b.

Kaiser Yamin, J. Tang, S. Cortes-Gomez, A. Sharma, Eric Horvitz, and Bryan Wilder. When agents say one thing and do another: Validating elicited beliefs from LLMs. *arXiv preprint arXiv:2602.06286*, 2026.

Yifan Zeng, Ojas Tendolkar, Raymond Baartmans, Qingyun Wu, Lizhong Chen, and Huaxiu Wang. LLM-RankFusion: Mitigating positional bias in large language model-based ranking. *arXiv preprint arXiv:2406.00231*, 2024.

Zhang, Zhang, Wu, Xu, and Gu. Beyond bradley-terry models: A general preference model for language model alignment. In *International Conference on Machine Learning (ICML)*, 2025. arXiv:2410.02197.

Jian-Qiao Zhu, Haijiang Yan, and Thomas L. Griffiths. Recovering event probabilities from large language model embeddings via axiomatic constraints. *arXiv preprint arXiv:2505.07883*, 2025.

A Where prior repair systems sit

Table 2 positions every system discussed in §2 against the four criteria that jointly define our claim: whether the repaired object is a choice the agent made for itself from a budget set (not a judgment about third-party items), whether the outcome measure is exogenous (no preference judgment enters it), whether the intervention is graded rather than binary, and whether the operator is a formal minimum-distance projection.

B Payoff exogeneity audit

Full detail of the four-part adversarial audit summarized in §3.3.

(1) No shared derived quantity beyond the exogenous (p, I). The payoff implementation has no dependency on the projection implementation; the optimal bundle x_t^* is a function of (p_t, I_t) alone and never references x_t or $\tilde{x}_t$. The one near-miss is that the projection's feasibility incumbent is a Cobb–Douglas demand *share-fitted to the agent's own observed data* — a plausible leakage channel on first read — but this quantity is never passed to the underlying MILP solver as an actual incumbent (the solver used has no such interface); it is used only to log an upper-bound sanity ceiling on the reported distance, never to steer the solution.

(2) The payoff's optimum is independent of revealed preference, checked empirically as well as by construction. Comparing raw (pre-repair) payoff between the 57 traces that were already

Table 2: Where prior repair systems sit. *Own choices*: the repaired object is a choice the agent made for itself from a budget set, not a judgment about third-party items. *Exogenous payoff*: an outcome into which no preference judgment enters.

	Own choices	Exogenous payoff	Graded dose	Min.-distance projection
POISE [Wang et al., 2026]	no	no	no	**yes**
Rationality layer [Chadwick et al., 2025]	no	no	no	**yes**
TrustJudge [Wang et al., 2025b]	no	no	no	no
CONSISTRE [Sun, 2026]	no	partial	no	no
LLM-RankFusion [Zeng et al., 2024]	no	**yes**	no	no (declined)
Innate preferences [Buchanan and Foster, 2026]	**yes**	no	no	no
GARP-EFM [Aguiar and Kashaev, 2026]	no	**yes**	no	no
HRC/DSPPO [Huang et al., 2026]	no	no	**yes**	no
Control vectors [Cook et al., 2026]	**yes**	no	**yes**	no
This paper	**yes**	**yes**	**yes**	**yes**

GARP-consistent and the 85 that were not: mean 0.7872 vs. 0.7899, Welch $t = -0.07$, $p = 0.94$. GARP-consistency status alone does not predict payoff, ruling out a population-level mechanical link between coherence and payoff score.

(3) Hand-derived mechanism check. Five traces spanning the full dose range (0.17 to 111.64) were inspected directly, comparing the direction of the actual projection movement $(\tilde{x} - x)$ against the direction toward the exogenous optimum $(x^* - x)$ via cosine similarity. Similarities were low (0.16–0.31) on the four traces where payoff improved after repair, and *negative* (-0.35) on the single trace where payoff worsened — the sign of the outcome tracks a geometric quantity the projection objective never references, evidence against the mechanism being a disguised optimization toward the payoff target.

(4) The severity-confound check. Larger-dose traces do start from worse raw payoff (Pearson $r = -0.41$, $p < 0.0001$), and worse-starting traces do have more room to improve ($r = -0.41$ between raw payoff and Δpayoff) — a genuine ceiling-effect confound. It does not explain the dose–response relationship away: the partial correlation of dose against Δpayoff controlling for raw payoff is $r = 0.784$ ($p = 7 \times 10^{-19}$, attenuated only slightly from the unconditional $r = 0.821$), and in a joint OLS regression the dose coefficient is highly significant ($t = 11.43$, $p < 10^{-16}$) while raw payoff's own coefficient is not ($t = -1.36$, $p = 0.18$).

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and §1 state exactly what is claimed (a measured dose–response relationship, an instrument-validity finding, and a replication of an independently published format effect) and what is not (novelty of repair, of projection, or of the sign of the effect); §6 states the design's scope limits explicitly.

   Guidelines:

   - The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
   - The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
   - The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
   - It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: §6 (Limitations) discusses CCEI's noise and power at 1.5B, the single-run-per-condition design, the two-model-family scope, the single fixed payoff weighting, and the adverse prior from three prior negative results.

   Guidelines:

   - The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
   - The authors are encouraged to create a separate "Limitations" section in their paper.
   - The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
   - The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
   - The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
   - The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
   - If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
   - While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. **Theory assumptions and proofs**

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [N/A]

Justification: The paper's contribution is a computational method (a MILP formulation for GARP-repair) building on a cited external representation theorem [Demuynck and Rehbeck, 2023]; it does not state or prove new theorems of its own.

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. **Experimental result reproducibility**

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: §4-5 specify the exact models, budget-set generator and seeding scheme, conditions, power calculation, retry protocol, and the full MILP formulation (§3) needed to reproduce the main results; the exogenous payoff is a closed-form function stated in full.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general. releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
  (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
  (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
  (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).

(d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

   Answer: [No]

   Justification: Code and raw per-trace logs are not released with this anonymized submission; both LLMs used are publicly available open-weight models and the full protocol (budget-set generator, seeds, conditions, retry rule, MILP formulation) is specified in the main text, so the experiment is reproducible from the paper alone even without the release.

   Guidelines:

   - The answer [N/A] means that paper does not include experiments requiring code.
   - Please see the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
   - While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
   - The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
   - The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
   - The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
   - At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
   - Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. **Experimental setting/details**

   Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

   Answer: [Yes]

   Justification: §4 specifies model identities, all three conditions' exact framing, the budget-set generator and its rejection rule, seed scheme, replicate count and power calculation, and the discard/retry protocol.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
   - The full details can be provided either with the code, in appendix, or as supplemental material.

7. **Experiment statistical significance**

   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

   Answer: [Yes]

Justification: Every reported comparison in §5 carries a p-value, and Table 1 and the dose–response statistics report means alongside sample sizes; CCEI comparisons additionally report standard deviations and, in §6, confidence-interval-based power limitations.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. **Experiments compute resources**

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: All inference was run locally on a single consumer CPU-class machine (no GPU, no commercial API calls), with the full $N = 30$ design measured at approximately 2.1 hours wall-clock (§4.1); every MILP projection solves in under 5 seconds with MIP gap $\leq 8.1 \times 10^{-5}$.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. **Code of ethics**

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?

Answer: [Yes]

Justification: The study elicits only synthetic, non-sensitive budget-allocation choices from open-weight language models; it involves no human subjects, no personal data, and no scraped content.

Guidelines:

- The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.

- If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. **Broader impacts**

    Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

    Answer: [Yes]

    Justification: See the Broader impacts paragraph preceding §7: the paper's own finding that 18% of repairs worsen the exogenous payoff is itself a caution against treating coherence-repair as a costless, universally beneficial deployment default.

    Guidelines:

    - The answer [N/A] means that there is no societal impact of the work performed.
    - If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
    - Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
    - The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
    - The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
    - If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. **Safeguards**

    Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

    Answer: [N/A]

    Justification: The paper releases no model or dataset; both LLMs used are existing, publicly released open-weight models used unmodified as measurement subjects.

    Guidelines:

    - The answer [N/A] means that the paper poses no such risks.
    - Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
    - Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
    - We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. **Licenses for existing assets**

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [Yes]

Justification: The two LLMs used are named with version in §4.1. Qwen2.5-1.5B-Instruct (Alibaba Cloud) is released under the Apache License 2.0. Llama-3.2-3B-Instruct (Meta Platforms, Inc.) is released under the Llama 3.2 Community License Agreement, a custom commercial license. Both are publicly available open-weight models used here for research purposes consistent with their license terms.

Guidelines:

- The answer [N/A] means that the paper does not use existing assets.
- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, `paperswithcode.com/datasets` has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. **New assets**

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [N/A]

Justification: This submission does not release a new dataset or model asset; the generated budget-set choice logs are experimental records, not a packaged new asset.

Guidelines:

- The answer [N/A] means that the paper does not release new assets.
- Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
- The paper should discuss whether and how consent was obtained from people whose asset is used.
- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. **Crowdsourcing and research with human subjects**

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [N/A]

Justification: The study involves no crowdsourcing and no human subjects; all choices are elicited from language models.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.

- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

    Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

    Answer: [N/A]

    Justification: The study involves no human subjects and required no IRB review.

    Guidelines:

    - The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
    - Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
    - We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
    - For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. **Declaration of LLM usage**

    Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

    Answer: [Yes]

    Justification: The two LLM agents under study are the paper's core experimental subject, not a writing aid: §4.1 names both models and their role as the choice-making agents whose revealed preferences are elicited, projected, and scored.

    Guidelines:

    - The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
    - Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.